



87 / 100

NODES ONLINE

01 ND-8091	02 ND-8092	03 ND-8093	04 ND-8094	05 ND-8095	06 ND-8096	07 ND-8097	08 ND-8098	09 ND-8099	10 ND-8100
11 ND-8101	12 ND-8102	13 ND-8103	14 ND-8104	15 ND-8105	16 ND-8106	17 ND-8107	18 ND-8108	19 ND-8109	20 ND-8110
21 ND-8111	22 ND-8112	23 ND-8113	24 ND-8114	25 ND-8115	26 ND-8116	27 ND-8117	28 ND-8118	29 ND-8119	30 ND-8120
31 ND-8121	32 ND-8122	33 ND-8123	34 ND-8124	35 ND-8125	36 ND-8126	37 ND-8127	38 ND-8128	39 ND-8129	40 ND-8130
41 ND-8131	42 ND-8132	43 ND-8133	44 ND-8134	45 ND-8135	46 ND-8136	47 ND-8137	48 ND-8138	49 ND-8139	50 ND-8140
51 ND-8141	52 ND-8142	53 ND-8143	54 ND-8144	55 ND-8145	56 ND-8146	57 ND-8147	58 ND-8148	59 ND-8149	60 ND-8150
61 ND-8151	62 ND-8152	63 ND-8153	64 ND-8154	65 ND-8155	66 ND-8156	67 ND-8157	68 ND-8158	69 ND-8159	70 ND-8160
71 ND-8161	72 ND-8162	73 ND-8163	74 ND-8164	75 ND-8165	76 ND-8166	77 ND-8167	78 ND-8168	79 ND-8169	80 ND-8170
81 ND-8171	82 ND-8172	83 ND-8173	84 ND-8174	85 ND-8175	86 ND-8176	87 ND-8177	88 ND-8178	89 ND-8179	90 ND-8180
91 ND-8181	92 ND-8182	93 ND-8183	94 ND-8184	95 ND-8185	96 ND-8186	97 ND-8187	98 ND-8188	99 ND-8189	100 ND-8190

74
HEALTHY

9
WARNING

4
FAULT

13
OFFLINE



ACTIVE ALARMS

13

- ND-8104

FAIL_OPEN Â· LHS Arrows

09:42:11
- ND-8137

FAIL_SHORT Â· Static Zone 1

09:39:02
- ND-8158

Cannot verify outputs

09:35:47
- ND-8181

FAIL_OPEN Â· RHS Arrows

09:33:20
- ND-8095

Commanded â‰‰ Actual

09:41:30
- ND-8113

Latched Â· SOLID-ALL

09:40:12
- ND-8121

Latched Â· SOLID-LEFT

09:38:55
- ND-8142

Commanded â‰‰ Actual

09:37:10
- ND-8093

Node OFFLINE

09:30:02

Search node ID or IP address

CONTROL SOURCE

governs all 100 nodes

REMOTE

SCADA · AUTO

LOCAL

HMI · MANUAL

Node ID	IP Address	Register	Status
ND-8091	10.20.7.11	40001	OK
ND-8092	10.20.7.12	40002	OK
ND-8093		40003	OFFLINE
ND-8094	10.20.7.14	40004	OK
ND-8095	10.20.7.15	40005	WARNING
ND-8096	10.20.7.16	40006	OK
ND-8097	10.20.7.17	40007	OK
ND-8098	10.20.7.18	40008	OK
ND-8099	10.20.7.19	40009	OK
ND-8100	10.20.7.20	40010	OK
ND-8101	10.20.7.21	40011	OK
ND-8102		40012	OFFLINE



< Signage List ● ND-8104

IP ADDRESS
10.20.7.24

REGISTER
40014

CONTROL SOURCE
LOCAL ● MANUAL

HARDWARE TELEMETRY

Connection Status

● ONLINE

Main Power

● OK

LHS Arrows

● FAIL_OPEN

RHS Arrows

● ON

Static Zone 1

● ON

Static Zone 2

● ON

Backup Battery

N/A ● NOT FITTED

MANUAL OVERRIDE CONTROL

● LOCAL (MANUAL) ● buttons live

commands ND-8104

COMMANDED
1 ● LHS CHASE

ACTUALLY EXECUTING
1 ● LHS CHASE

● IN SYNC

OPERATOR MODES

0
Arrows OFF

1
LHS Chase

2
RHS Chase

TECHNICIAN MODES ● LATCH UNTIL CHANGED

3
Both Chase

4
Solid Left

5
Solid Right

6
Solid All

Per-LED Control

IP ADDRESS

10.20.7.24

REGISTER

40014

CONTROL SOURCE

LOCAL • MANUAL

HARDWARE TELEMETRY

Connection Status

Main Power

LHS Arrows

RHS Arrows

Static Zone 1

Static Zone 2

MANUAL OVERRIDE CONTROL

commands ND-8104

LLY EXECUTING

LHS CHASE

IN SYNC

se

2

RHS Chase

5

Solid Right

6

Solid All

Per-LED Control

ND-8104 • direct MOSFET override • raw bitmask

Technician-level. Individual LEDs stay latched in a non-standard state until a mode is re-commanded.

< LEFT ARROW • TAP AN LED

1

2

3

4

5

> RIGHT ARROW • TAP AN LED

1

2

3

4

5

ALL LEDs ON

ALL OFF

RAW VALUE

10101 01101

10685

Cancel

Apply

Backup Battery

N/A • NOT FITTED

Per-LED Control

GATEWAY & SERVICES

MQTT Broker (Paho)

●

CONNECTED

Modbus TCP Server

●

RUNNING

502

Master Gateway IP

127.0.0.1

Gateway Service

●

ACTIVE

Last Telemetry Sync

09:42:07

NETWORK HEALTH

87

ONLINE NODES

13

OFFLINE

■

9

WARNING

■

4

IN FAULT

TELEMETRY REFRESH

Asks every node to re-publish its full telemetry set.
This is a refresh, not a fail-safe – nodes hold their
last command when the network drops.

BROADCAST PING